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# Security-enhanced optical nonlinear cryptosystem based on phase-truncated Fourier transform

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**Abstract**
In this paper, an optical nonlinear cryptosystem based on phase-truncated Fourier transform is proposed. A time-varying nonlinear term, denoted as $\mu(x_1, y_1, t)$, is introduced within the framework of the classical PTFT cryptosystem to enhance the security. This term, characterized by its randomly generated complex distribution, remains spatially separated from the input image in the input plane and has no contribution in the decryption of the input image. The proposed encryption approach capitalizes on the remaining information capacity inherent in the optical system and enhances security. The proposed enciphering scheme is proved to be resistant to various attacks, including ciphertext-only attacks, known-plaintext attacks and chosen-plaintext attacks. In addition, the statistical analysis and comparison with the existing schemes has also been carriedout to demonstrate its strengths.

**Keywords** Optical encryption · Fourier transform · Nonlinear cryptosystem · Security analysis

## 1 Introduction

Optical image encryption has been a preferred research area for more than a decade due to its unmatchable intrinsic advantages of fast parallel processing and multidimensional encryption capabilities (Javidi et al. 2016). Optical image encryption started with the Fourier transform-based double random phase encoding (*DRPE*) system (Refregier and Javidi 1995) in 1995. *DRPE* system used two statistically independent random phase masks in the input and output plane to convert an input image into stationary white noise. Ever since the *DRPE* cryptosystem was proposed, plenty of research based on DRPE has been carried out (Goudail et al. 1998; Wang et al. 2000; Monaghan et al. 2007; Xin et al. 2007; Situ et al. 2008; Chen and Chen 2014; Fatima and Nishchal 2016;

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Rajput and Nishchal 2013a, 2013b). Contributions comprising experimental outcomes have also been presented by researchers (Gaur et al. 2023; Dou et al. 2019; Anjana et al. 2021; Zea et al. 2017, 2016; Shen et al. 2016; Barrera et al. 2014a, 2014b). The *DRPE* system has been implemented in other domains as well, such as Fresnel transform domain (Situ and Zhang 2004), the fractional Fourier transform domain (Unnikrishnan et al. 2000; Unnikrishnan and Singh 2000; Jin and Yan 2007; Tao et al. 2007; Liu et al. 2013), and Gyrator transform domain (Liu et al. 2010; Abuturab 2012).

For a cryptosystem to be secure, it is important that one should not be able to retrieve the input image from its cipher image without knowing the correct key. Although optical cryptosystems have multidimensional encryption capabilities and large key space, implemented transformations, such as Fourier transform, are linear, and this leads to the weakness of the *DRPE* cryptosystem to the various attacks (Liu et al. 2015; Guo et al. 2016), such as known-plaintext attack (*KPA*) (Peng et al. 2006a; Qin et al. 2010; Tashima et al. 2010; Gopinathan et al. 2006; Liu et al. 2009; Frauel et al. 2007) and chosen-plaintext attack (CPA) (Carnicer et al. 2005; Frauel et al. 2005; Peng et al. 2006b).

To overcome the weakness of *DRPE* based cryptosystems posed by their linear structure. Peng et al. in 2010, proposed a cryptosystem which was based on phase truncation operation to give different keys for encryption and decryption but the keys were not publically available for encryption (Qin and Peng 2010). Since both the keys were different they termed their cryptosystem as asymmetric. Though, it is not a public key cryptography, the term asymmetric is used just to emphasis the fact that keys for encryption and decryption are different. Later on, the researchers started using this terminology. The *PTFT* based cryptosystem proposed by them overcomes the weakness of *DRPE*, but later, it was shown to be susceptible to interception by a phase retrieval technique with single intensity measurement. The input image is obtained from its corresponding cipher image via phase retrieval algorithms such as Gerchberg-Saxton (*GS*) algorithm and the hybrid input–output (HIO) algorithm (Gerchberg 1972; Fienup 1982).

To strengthen the classical *DRPE* cryptosystem, a security-enhanced nonlinear cryptosystem using a time-varying nonlinear term was designed in 2020 by Dou et al. (2020). Although the security evaluation of a nonlinear cryptosystem by Singh et al. (2021) shows its vulnerability to modified CPA and KPA attacks. Later on, this nonlinear cryptosystem is improved by using a modified GS iterative algorithm in the Fourier domain by Girija et al. (2021).

Furthermore, recent advancements in image encryption have led to the exploration of new techniques that enhance security and robustness. A quantum logistic map combined with a hyperchaotic system has been proposed for noise reduction in encrypted images (Mohamed et al. 2022), while a multiple color image fusion, compression, and encryption scheme utilizing compressive sensing, chaotic-biometric keys, and optical fractional Fourier transform has been introduced (Abuturab and Alfalou 2022). An asymmetric double image encryption, compression, and watermarking scheme based on orthogonal-triangular decomposition with column pivoting has also been developed (Anjana et al. 2022). Additionally, an optical asymmetric single-channel cryptosystem based on QZ synthesis for color images has been proposed, achieving enhanced security against special attacks (Shen et al. 2022). Moreover, a multiple color image cryptosystem based on coupled-logistic-map-biometric keys, QR decomposition with column pivoting, and optical Fresnel transform has been presented (Abuturab 2023). Lastly, a fingerprint image encryption technique utilizing phase retrieval algorithms in the gyrator wavelet transform domain has been introduced (Mehra and Nishchal 2023).

In this paper, the time-varying nonlinear term is introduced in classical *PTFT* based cryptosystem to obtain a security-enhanced nonlinear encryption mechanism. The security evaluation of the proposed cryptosystem reveals its robustness against classical attacks, as well as resistance to noise and occlusion attacks.

## 2 Nonlinear cryptosystem based on *PTFT*

The conventional *PTFT* cryptosystem designed by Qin and Peng (2010) is shown in Fig. 1. The *PTFT* system uses two random phase masks, $RPM1$ and $RPM2$, as encryption keys.

The following steps show the complete process of encryption. $FT$ represents the Fourier transform operation. Here, $PT$ and $PR$ denote phase truncation and phase reservation operations, respectively. In $PT$ operation, the phase part is discarded, and the magnitude part is preserved, whereas in the $PR$ operation, the magnitude part is discarded, and the phase part is preserved, which yields decryption keys $P1$ and $P2$.

The intensity distribution $I(u, v)$ of the input image $I(x, y)$ in the Fourier domain is expressed as:

$$I(u, v) = |I(u, v)| \times exp(i\pi\varphi(u, v)) \tag{1}$$

where $|I(u, v)|$ is the magnitude of $I(u, v)$ and $exp(i\pi\varphi(u, v))$ is its phase.

Ciphertext $E(x, y)$ is obtained from the following:

$$h(u, v) = PT\left[FT(I(x, y) \times RPM1)\right] \tag{2}$$

$$E(x, y) = PT[IFT(h(u, v) \times RPM2)] \tag{3}$$

The decryption keys $P1(u, v)$ and $P2(x, y)$ are obtained as:

$$P1(u, v) = PR\left[FT(I(x, y) \times RPM1)\right] \tag{4}$$

$$P2(x, y) = PR[IFT(h(u, v) \times RPM2)] \tag{5}$$

The decryption process is shown in Fig. 2. The decryption keys $P1(u, v)$ and $P2(x, y)$ are used to get the original image $I(x, y)$ from the following equations:

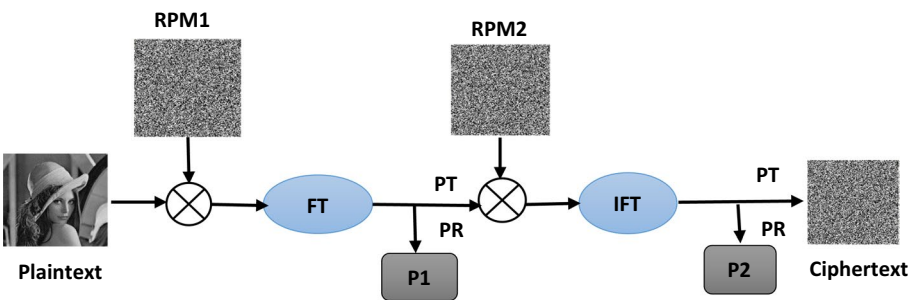



**Fig. 1** Encryption process of the *PTFT* cryptosystem

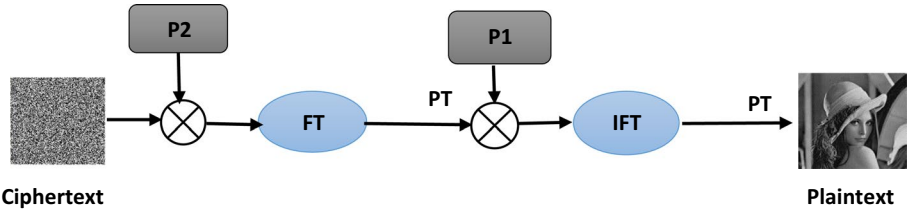



**Fig. 2** Decryption process of the *PTFT* cryptosystem

$$h(u,v) = PT\left[FT(E(x,y) \times P2(x,y))\right] \tag{6}$$

$$I(x,y) = PT[IFT(h(u,v) \times P1(u,v))] \tag{7}$$

The $RPM1$ and $RPM2$ are the keys used at the encryption side for generating a cipher image, whereas $P1(u,v)$ and $P2(x,y)$ serve as decryption keys at the receiver end for decoding the encrypted image.

The PTFT cryptosystem is based on a two-dimensional quadratic phase system. The space-bandwidth product (SBP) of the optical cryptosystem to calculate the information capacity is measured by the following equation:

$$S_{ptft} = \frac{l_{x_1} l_{x_2}}{f\lambda} \times \frac{l_{y_1} l_{y_2}}{f\lambda} \tag{8}$$

Here, $l_{x_1} \times l_{y_1}$ is the area of input image in the input plane and $l_{x_2} \times l_{y_2}$ is the area for the phase mask in Fourier plane. $f$ is the focal length of the lenses used, $S_{ptft}$ number gives the maximum number of input image pixels that can be encrypted by the cryptosystem. In *PTFT* and *DRPE* based cryptosystems, the information capacity can be fully occupied by the input image. In other words, if $S_{ptft}$ is $m \times n$ pixels, an input image with a maximum size of $m \times n$ pixels can be encrypted. However, in most cases, the size of the input image is generally smaller than $m \times n$ pixels. Therefore, the remaining information capacity can be utilized in encryption by introducing a randomly generated time-varying nonlinear complex term $\mu(x_1, y_1, t)$. This term is located spatially separated from the input image in the input plane. We have proposed a nonlinear cryptosystem with the same optical implementation as that of PTFT based cryptosystem using this nonlinear term $\mu(x_1, y_1, t)$.

Figure 3 illustrates the input plane of the nonlinear cryptosystem with space-bandwidth product ($SBP$) $400 \times 400$ pixels and the original image with the size of $f(x_1, y_1)$ is bonded with phase key $R(x_1, y_1)$. Here, $\mu(x_1, y_1, t)$ is the time-varying nonlinear term introduced to occupy the remaining information capacity surrounding the input image.

Therefore, the encrypted image in nonlinear *PTFT* based cryptosystem is obtained by the following steps:

$$\begin{aligned} E(x,y) &= \left|\left\{\mathrm{IFT}\left[\left\{\left|\left(FT\left\{(I(x,y) \times RPM1) + \mu(x_1,y_1,t)\right\}\right|\right)\right\} \times RPM2\right]\right\}\right| \\ &= \left|\mathrm{IFT}\left[\left\{\left|FT\left\{(I(x,y) \times RPM1) + \mu(x_1,y_1,t)\right\}\right|\right\} \times RPM2\right]\right| \\ &= |IFT|FT\{(I(x,y) \times RPM1)\}| \times IFT(RPM2)| + \mu(x_1,y_1,t) \times |IFT(RPM2)| \end{aligned} \tag{9}$$

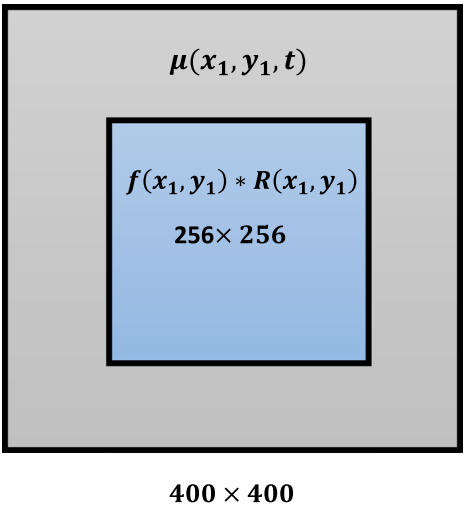



**Fig. 3** Input plane of nonlinear *PTFT* cryptosystem: the size of original image $f(x_1, y_1)$ bonded with RPM $R(x_1, y_1)$ is 256 ×256 pixels; $\mu(x_1, y_1, t)$ is introduced as the nonlinear term, which is of the size of *SBP* of the cryptosystem

Equation (9) shows that the nonlinear term is spatially separated from the image bonded with RPM in the input plane. Therefore, it does not contribute to the decryption of the image and is not required to be transmitted over the channel. This means that the nonlinear term is not known to anyone. This increases the security of the cryptosystem without the additional cost of management and distribution of nonlinear term. Decryption involves the same process as shown in Fig. 2.

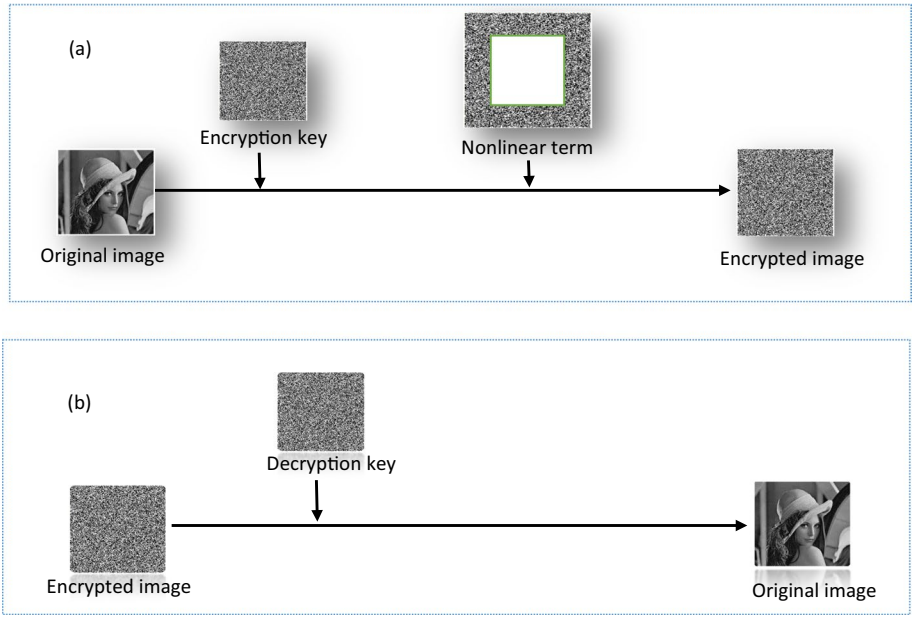



**Fig. 4** Encryption and decryption algorithms of nonlinear *PTFT* cryptosystem

Figure 4 shows the diagrammatic representation of the nonlinear *PTFT* cryptosystem. This algorithm can also be implemented digitally, wherein the time-varying nonlinear term $\mu(x_1, y_1, t)$ is randomly generated. The optical implementation of nonlinear *PTFT* is the same as that of conventional *PTFT* cryptosystem. The nonlinear term $\mu(x_1, y_1, t)$ and bonded pair $I(x, y) \times RPM1$ can be uploaded to spatial light modulators (SPMs) to carry out the encryption in an optical setup.

## 3 Results and discussion

The numerical simulations and various outputs for the proposed cryptosystem have been obtained using *MATLAB* 2022a. The space-bandwidth product (*SBP*) of the proposed cryptosystem has been set to 400 × 400 pixels. Figure 5 shows the validation results for the scheme: (a) the input Lena image of size 256 × 256 pixels, (b) one of the RPMs of size 256 × 256 pixels used in encryption, (c) the randomly generated time-varying

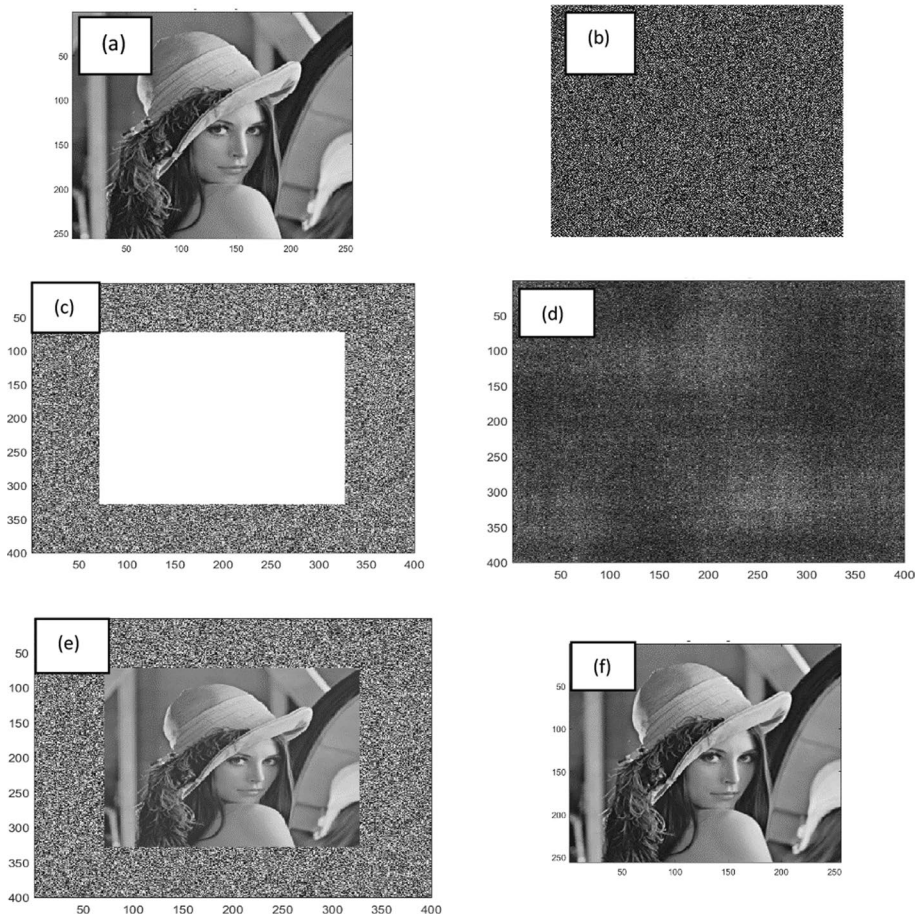



**Fig. 5** Validation results for the proposed cryptosystem: **a** input image, **b** RPM1, **c** nonlinear term $\mu(x_1, y_1, t)$, **d** encrypted image, **e** decrypted image with $\mu(x_1, y_1, t)$, **f** decrypted original image

nonlinear term $\mu(x_1, y_1, t)$ of size 400 × 400 pixels, (d) the encrypted image, (e) the decrypted image surrounded by the nonlinear term, and (f) the extracted original image. The results obtained for encryption and decryption validate the proposed scheme.

A cryptosystem is considered secure if it is sensitive to its keys. To check the key sensitivity of the proposed scheme, we have obtained decryption results by altering the keys in the decryption process. Figure 6 shows the decrypted image when (a) keys key1 and key2 are interchanged, (b) key1 is used as both the decrypting keys, (c) key2 is used as both the decrypting keys, (d) key2 was correct but key1 was incorrect, (e) key1 was correct, but key2 was incorrect. Key sensitivity analysis shows that key1 has greater

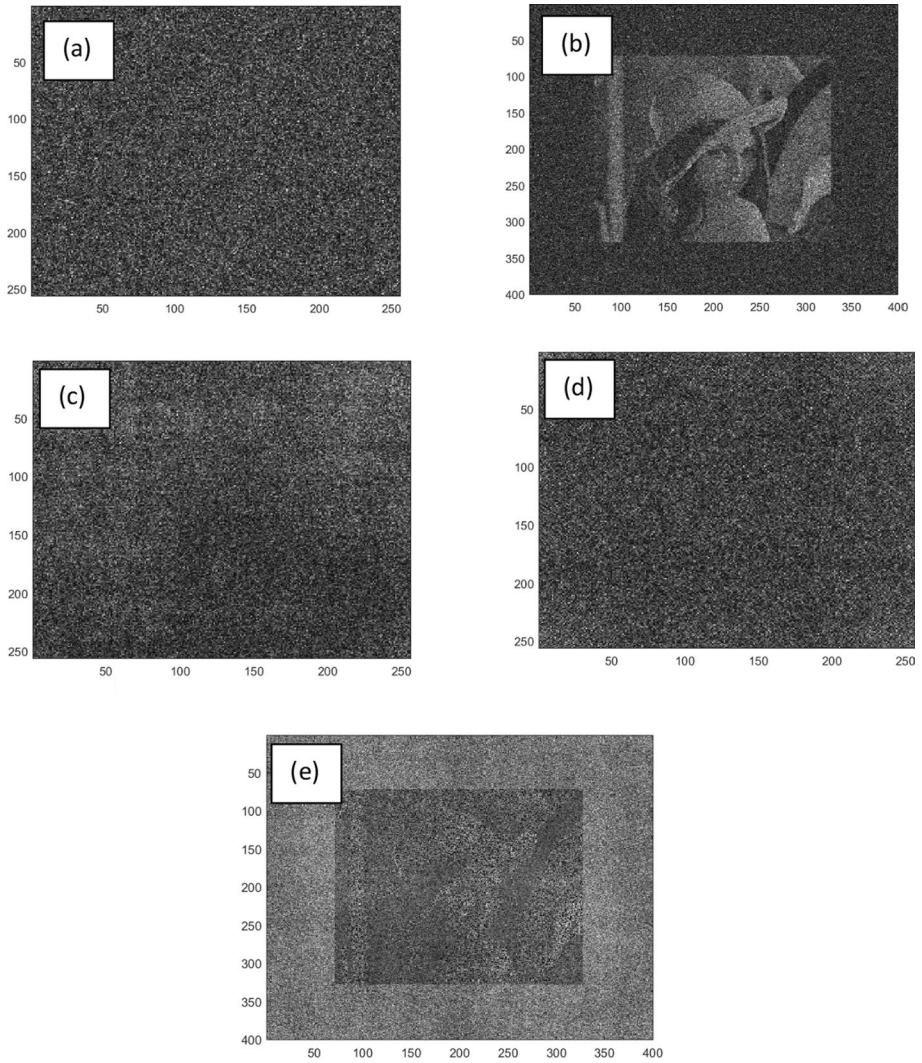



**Fig. 6** Key sensitivity analysis: **a** key1 and key2 interchanged, **b** key1 is used as both the decrypting keys, **c** key2 is used as both the decrypting keys, **d** correct key2 but incorrect key 1, **e** correct key1 but incorrect key2

**Table 1** Statistical parameters of the proposed scheme

| Parameters | SSIM | Entropy | Execution time | Correlation -Coefficient | MSE | PSNR |
|---|---|---|---|---|---|---|
| Proposed scheme | 1 | 7.7610 | 0.6 s | 1 | 4.55e − 32 | 313.4155 dB |

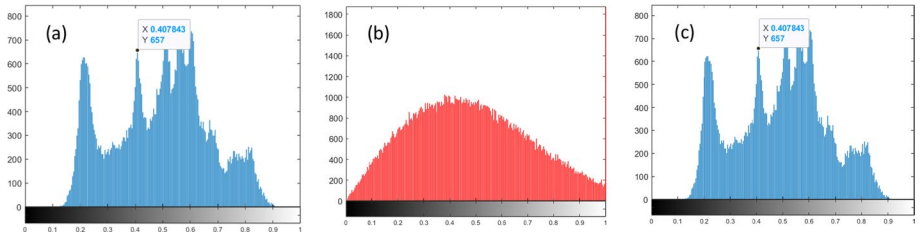



**Fig. 7** Histogram analysis of the proposed scheme: histograms of **a** input image Lena, **b** encrypted image, and **c** decrypted image

significance in decryption as compared to key2, as applying only key1 results in a recognizable decrypted image.

Statistical parameters are also evaluated for the proposed scheme, which is shown in Table 1. Table 1 shows the *SSIM* and correlation coefficient between the input and output images, entropy value of the encrypted image, time taken during the encryption, *MSE*, and *PSNR* values between input and output images. *SSIM* and *CC* values indicate the similarity between the two images. Here, both the metrics have values equal to 1, which demonstrates that the images have been faithfully recovered through decryption. The Entropy of the encrypted image is 7.7, which indicates the high randomness in the encrypted image.

Lesser execution time together with lower *MSE* and high *PSNR* values indicate the strengths of the proposed cryptosystem.

Figure 7 shows the histograms of the (a) input image Lena, (b) encrypted image, and (c) decrypted image, respectively. As can be seen, the histograms of the input image and decrypted image are same, implying the decryption has been achieved successfully, whereas the histograms of the enciphered image is uniform implying that the encrypted image is completely random.

Figure 8 shows the noise attack results when salt & pepper noise and Gaussian noise of varying densities are applied on the encrypted image. Figure 8 shows the decrypted image with salt and pepper noise of densities (a) 0.25, (b) 0.5 and (c) 0.8, respectively. At the same time, Fig. 8d and e shows the results obtained by introducing Gaussian noise with zero mean and standard deviation (d) 0.25, (e) 0.5 and (f) 0.8 values. Clearly, the proposed encryption scheme has endured both attacks. Figure 9 shows the *CC* vs noise density plots.

Occlusion of the encrypted image can occur during transmission; therefore, it is necessary to have an occlusion attack-resistant cryptosystem. Figures 10 and 11 show the results obtained by occluding the encrypted image (a) by 20% and (c) 60%; (b) and (d) are the corresponding decrypted images. Figure 11 shows the *CC* vs. occluded area plot.

The proposed scheme is non-linear in nature. Based on the encryption scheme's intrinsic non-linear nature, it gains resilience against basic attacks including ciphertext-only attacks (where only the scheme and the encrypted message are available), known-plaintext attacks (where both the scheme and a plaintext-ciphertext pair are accessible), and

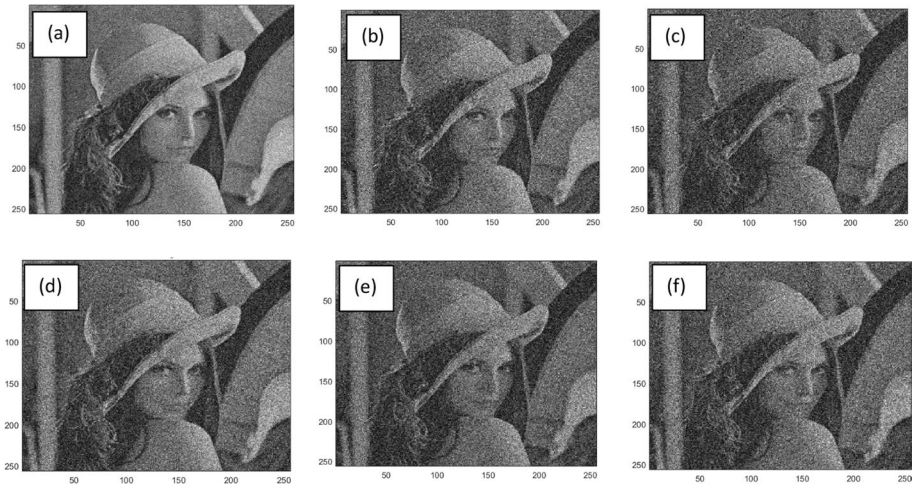



**Fig. 8** Noise attack analysis of the proposed scheme

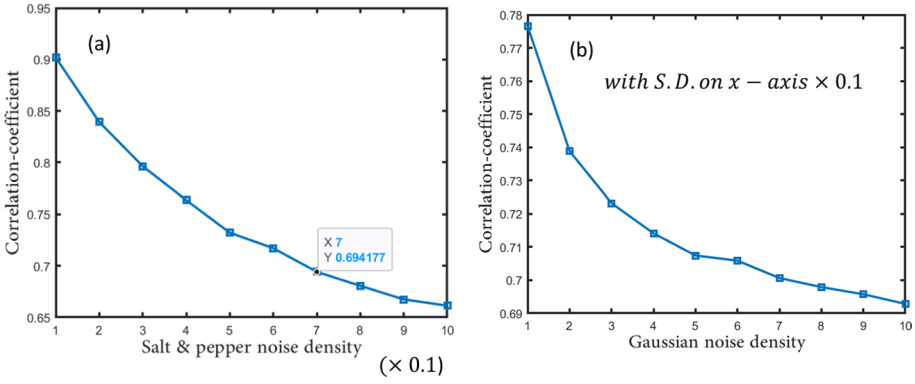



**Fig. 9** Noise density vs *CC* curve for the proposed scheme

chosen-plaintext attacks (where the scheme, a personally chosen plaintext, and its matching ciphertext are accessible). It is worth mentioning that the keys generated by any of the above attacks are of no use as private keys depend on the plaintext. With different plaintext, the decryption keys necessary for the decryption process also change. In simpler terms, the decryption keys utilized in such a system act as one-time pads and therefore the scheme is resistant to traditional attacks. The outcome of the Known Plaintext Attack (KPA) on the proposed scheme is given in Fig. 12.

The Known Plaintext Attack (KPA) is a cryptanalytic technique where an attacker possesses both the plaintext-ciphertext pairs from a given encryption scheme. In this attack, the attacker uses the known plaintext to retrieve the corresponding decryption keys and then attempts to decrypt other ciphertexts using these keys. To evaluate the scheme's resistance, we conducted a KPA experiment by employing the Barbara image as the known plaintext. The derived decryption keys from the Barbara image were then employed to decipher the ciphertext obtained from the Lena image. Notably, the decryption process did

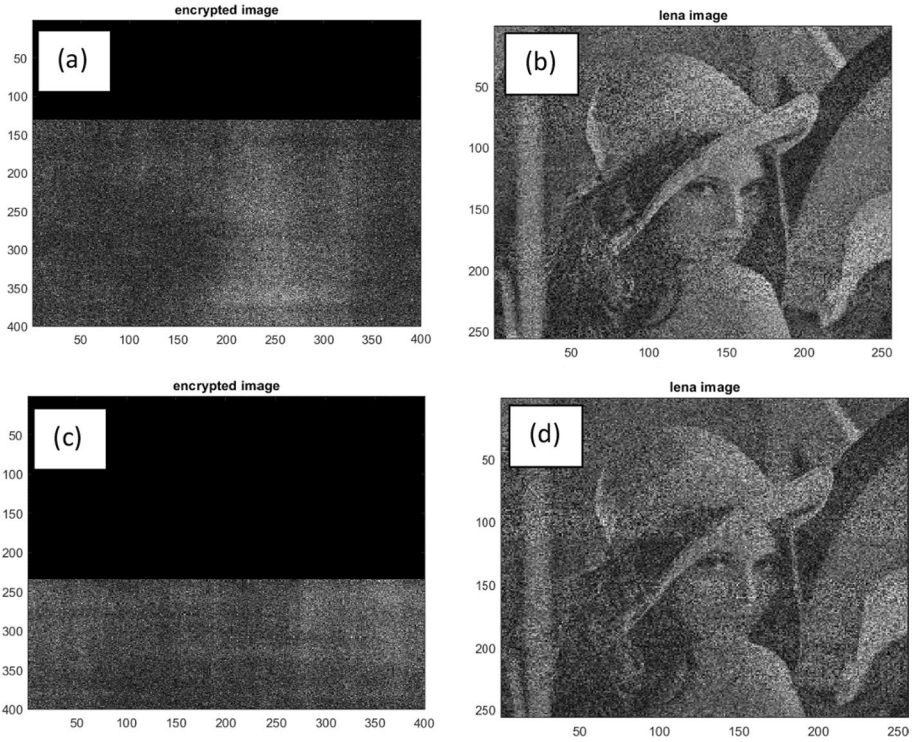



**Fig.10** Occlusion attack analysis for the proposed scheme

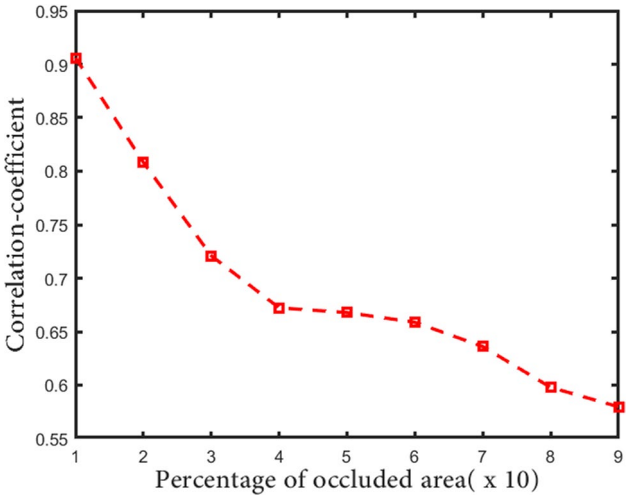



**Fig.11** Occluded area vs cc curve for the proposed scheme



**Fig.12** Known-plaintext attack output on the Lena ciphertext

not yield the expected plaintext Lena image, confirming the scheme's robustness against KPA.

A comprehensive comparative analysis has also been conducted, juxtaposing the proposed schemes with those by Anjana et al. (2022), Gong et al. (2018), Singh (2018), Xu et al. (2019), Gong et al. (2019), Yang et al. (2019). The comparison is based on factors such as the utilized transform domain, the approach employed, substitution and permutation techniques utilized, encryption duration, PSNR of decrypted image, and resilience against attacks. The summarized results of this analysis are presented in Table 2 below:

In light of the comparative analysis presented in Table 2, it becomes evident that the proposed scheme outperforms its counterparts in several crucial aspects. The careful evaluation across parameters such as the transform domain utilized, applied approach, substitution and permutation techniques employed, encryption period, PSNR of decrypted image, and resilience against attacks offers a comprehensive picture of the scheme's capabilities. Particularly noteworthy is its robustness against a range of attacks, including occlusion, noise, and known/chosen plaintext attacks. Moreover, the efficiency of the proposed scheme, as indicated by its relatively shorter encryption period and high PSNR values, further underscores its superiority. This analysis collectively underscores the potential significance and effectiveness of the proposed scheme in secure image encryption applications.

## 4 Summary

An optical and digital nonlinear cryptosystem based on phase-truncated Fourier transform has been proposed. A time-varying nonlinear term is introduced in classical *PTFT* asymmetric cryptosystem to enhance the security. The nonlinear term is a randomly generated complex distribution and has no contribution in the decryption of the input image. The cryptosystem has been analyzed by observing statistical parameters, key alterations and noise contaminations. Here, we have also verified the cryptosystem's strength against the classical cryptographic attacks and compared the proposed scheme with the existing schemes. Upon comparison with the existing schemes, the proposed encryption algorithm consistently demonstrated commendable performance across all the aforementioned metrics.

**Table 2** Comparison analysis for the proposed scheme

| Parameters | L. Gong et al. (Gong et al. 2018) | H. Singh scheme (Singh 2018) | Q. Xu et al. (Xu et al. 2019) | L. Gong et al. (Gong et al. 2019) | Y. et al. (Yang et al. 2019) | S Anjana et al. (Anjana et al. 2022) | Proposed Scheme |
|---|---|---|---|---|---|---|---|
| Transform domain utilized | Discrete fractional random transform | Gyrator | Discrete wavelet domain | Spatial | Spatial | Fresnel transform | Fourier domain |
| Applied approach | Symmetric | Asymmetric | Symmetric | Symmetric | Symmetric | Asymmetric | Asymmetric |
| Substitution techniques | √ | √ | √ | √ | √ | √ | √ |
| Permutation techniques | Hyper chaotic system | × | 2D-SLIM Hypercha-otic map | Arnold transform & a chaotic system | Fractional-order Chen hyper-chaotic system | × | × |
| Encryption period (Seconds) | – | – | 0.05 s for CR = 0.75 | > 6 s for image size $256 \times 256$ (increases with image size) | – | 0.43 s for encryption 0.52 s for compres-sion | 0.64 s |
| PSNR of decrypted Image | 76.56 dB | 310.9292 dB | 29 dB | > 30 dB | 35.8548 dB | 634 dB | 313.415 dB |
| Performance against attacks | Robust against noise attack but can with-stand a small degree of occlusion | Robust against occlusion and noise attack to some degree | Robust against known/ chosen plaintext, low-intensity noise and minor occlusion | Robust against noise attack but can with-stand a small degree of occlusion | Robust against noise attack and occlusion attack | Robust encryption and resistant against occlusion attack | Robust encryption and resistant against occlusion attack, noise attack, known/ chosen plaintext attack |

**Author contributions** "All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by SA, KSG, HS, PS, and AKY. The first draft of the manuscript was written by SA, and KSG and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript."

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**Research data policy and data availability** At The NorthCap University, we recognize the importance of open and transparent science. We encourage the sharing of research data to foster scientific collaboration and reproducibility. In accordance with this commitment, The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

## Declarations

**Conflict of interest** "The authors have no relevant financial or non-financial interests to disclose."

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